ABEEHA BAIG

AI/ML Engineer | Generative AI, LLMs, AI Agents, RAG, Computer Vision

Karachi, Pakistan

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PROFILE

AI/ML Engineer with ~3 years of experience building and deploying production Machine Learning and Generative AI systems in healthcare, pharma, and fintech. Builds AI agents (LangChain, LangGraph), advanced RAG pipelines, LLM applications, and computer vision/OCR solutions, shipped as FastAPI/Flask services on GCP (Vertex AI, Cloud Run) and AWS with Docker and CI/CD. Seeking English-language AI/ML roles in Germany (Berlin, Munich or remote-first teams).

WORK EXPERIENCE

AI Innovation Engineer | *Qordata – Karachi, Pakistan* July 2025 – Present

- Architected Agentic RAG workflows for real-time pharmaceutical compliance monitoring, automating web search and comparison of drug data.
- Built agentic chatbot solutions with LangChain and LangGraph that support decision-making on product changes, and developed SQL agents that let the chatbot query structured data in natural language.
- Integrated Gemini 3 Flash and Gemini 3.1 Flash Lite as LLM back-ends for scenario-based solutions, using structured prompt engineering for reliable, production-ready outputs and LLM-based rule classification.
- Engineered document OCR pipelines with AWS Textract, PaddleOCR, RapidOCR, and Google Gemini DocumentAI for handwritten and recursive text extraction.
- Trained YOLOv8 and YOLOv9 object detection models to learn image patterns and detect circles in documents.
- Implemented image preprocessing (adaptive thresholding, deskewing, sharpening) to improve OCR input quality and extraction accuracy.
- Deployed containerized AI services and data pipelines on GCP (Cloud Run, Cloud Storage, Firestore, Vertex AI) and AWS, with CI/CD for end-to-end delivery.
- Developed front-end UIs in TypeScript and React.js for AI applications.
- Used AI coding assistants (GitHub Copilot, Antigravity) to speed up development and evaluated emerging AI developer tooling.

Machine Learning Engineer | *PixelEdge Solutions (Pvt.) Limited – Karachi, Pakistan* Oct 2023 – June 2025

- Designed and deployed LangChain AI agents that handle multi-step queries across healthcare and financial products.
- Built modular, basic, and advanced RAG pipelines with HyDE, step-back prompting, vector databases, and re-ranking to improve chatbot retrieval quality and relevance.
- Benchmarked multiple vector databases, comparing HNSW (approximate nearest neighbor) and flat (exact) indexing for retrieval performance.
- Developed SQL Agent and Pandas Agent tools with LangChain (Python REPL) for fintech products, letting LLMs query and reason over financial datasets and return natural-language insights.
- Built financial data extraction and processing pipelines using regex, NLP techniques, and Pandas for transaction analytics, generating formatted Excel outputs.
- Automated web scraping with BeautifulSoup and Selenium to collect data for NLP and analytics pipelines.
- Fine-tuned GPT-2 on healthcare data for a pharmaceutical project to deliver domain-specific next-word prediction.
- Built project audit summarization with LLMs using prompt engineering techniques.
- Applied DBSCAN clustering and sentiment analysis to customer reviews for customer segmentation and insight extraction.
- Integrated and prompt-engineered multiple LLMs (OpenAI, Gemini, Claude, DeepSeek, LLaMA) to solve complex, domain-specific problems in healthcare and finance.
- Developed and shipped Flask and FastAPI REST APIs with robust error handling for development and production environments.
- Deployed and managed projects on AWS cloud services, covering deployment and ongoing infrastructure management.

Machine Learning Intern | *Techno-Hacks Edu Tech – Karachi, Pakistan* Aug 2023 – Sep 2023

- Built classical machine learning models for classification, prediction, and regression on real-world datasets, applying core ML principles and predictive analysis techniques.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn; classification, regression, clustering (DBSCAN), predictive modeling, model training, fine-tuning (GPT-2)

Generative AI & LLMs: OpenAI, Gemini, Claude, DeepSeek, LLaMA; prompt engineering, structured outputs, tool/function calling, LLM fine-tuning

AI Agents & Frameworks: LangChain, LangGraph, AI agents, multi-agent/agentic workflows, SQL Agent, Pandas Agent, Agentic RAG

RAG & Retrieval: Retrieval-Augmented Generation (modular, advanced), vector databases, embeddings, semantic search, HNSW & flat (exact) indexing, re-ranking, HyDE, step-back prompting

Computer Vision & OCR: YOLOv8, YOLOv9, object detection, image preprocessing (deskewing, adaptive thresholding, sharpening), OCR (AWS Textract, PaddleOCR, RapidOCR, Gemini DocumentAI), image recognition

NLP: Sentiment analysis, text summarization, text preprocessing, regex, information extraction

Data Engineering: Pandas, NumPy, web scraping (BeautifulSoup, Selenium), ETL/data pipelines, Excel reporting

Backend & Frontend: FastAPI, Flask, REST APIs; React.js, TypeScript

Cloud & MLOps: GCP (Vertex AI, Cloud Run, Cloud Storage, Firestore), AWS, Docker, containerized deployments, CI/CD, Git/GitHub

Tools: Streamlit, GitHub Copilot, Antigravity (AI coding assistants), n8n

EDUCATION

Master of Science – Data Science | NEDUET — Jan 2025 – Present | CGPA: 3.33/4.00

Bachelor of Engineering – Avionics | KIET — Jan 2020 – Dec 2023 | CGPA: 3.56/4.00

University projects: Python programming projects and image recognition systems.

LANGUAGES

English: C1 — professional / working language

COURSES & CERTIFICATIONS

- Generative AI with LLMs — Coursera
- N8n for automation – n8n academy
- AI Agents — Hugging Face
- OCI Generative AI Professional Certification — Oracle University
- Building Recommendation Systems using Machine Learning — Great Learning Academy
- Building Systems with ChatGPT API and Gemini API — DeepLearning.AI
- Python Programming — Open Weaver

PERSONAL PROJECTS

AI Health & Macros Management Agent (LangGraph) *(in progress)*
Healthcare AI agent built with LangGraph to manage health goals and nutrition/macro tracking.

Modular RAG-Based Chatbot for Oracle Database Security Q&A — *github.com/abeeha-baig/RAG-ODS*
Retrieval-Augmented Generation Q&A system (Streamlit + LangChain) that retrieves from embedded PDF documents to generate coherent, grounded conversational answers.

Financial Data Bot Using LangChain's Pandas Agent — *github.com/abeeha-baig/financial-chatbot*
Chatbot using OpenAI LLMs and the LangChain Pandas Agent for natural-language querying of CSV-based financial datasets.

Lifestyle Agent Using LangChain and FireCrawl — *github.com/abeeha-baig/Lifestyle-coach-agent*
Wellness/productivity agent that scrapes articles (Medium, Healthline) via Firecrawl and summarizes them with GPT-4o-mini; built with Python, Streamlit, LangChain, and secure API-key handling.

Next-Word Prediction via Fine-Tuned GPT-2 — *github.com/abeeha-baig/Next-Word-Prediction*
Fine-tuned GPT-2 on healthcare data to build a domain-specific next-word prediction model.